4-23

4–23 An ideal gas undergoes two processes in a piston–cylinder device as follows:

1–2 Polytropic compression from T_1 and P_1 with a polytropic exponent n and a compression ratio of $r = V_1/V_2$.

2–3 Constant pressure expansion at $P_3 = P_2$ until $V_3 = V_1$.

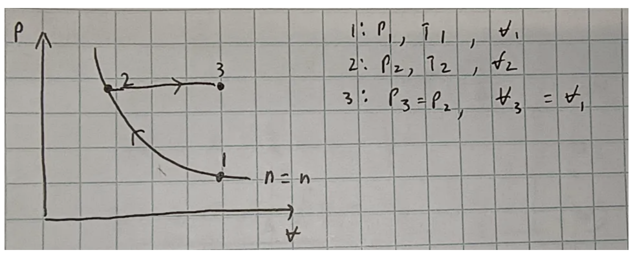
(*a*) Sketch the processes on a single *P*-*V* diagram.

(*b*) Obtain an expression for the ratio of the compression-to-expansion work as a function of n and r.

(*c*) Find the value of this ratio for values of $n = 1.4$ and $r = 6$.

Answers: (b) $\frac{1}{n-1}\left(\frac{1-r^{1-n}}{r-1}\right)$ (c) 0.256

a) A polytropic process describes a process where PV^n = constant (this also applies to specific volume Pv^n = constant since the mass is a constant). In this case of an ideal gas in a cylinder, n must be positive since we would expect pressure to increase as volume decreases, a negative n would imply the opposite.



b) A definition of work done by a system is

$$W = \int P\, dV$$

Where in a polytropic process where $n \neq 1$,

$$PV^n = C \implies P = CV^{-n}$$

$$W = \int_1^2 CV^{-n}\, dV = \boxed{\frac{C}{-n+1}\left(V_2^{-n+1} - V_1^{-n+1}\right)}$$

For the case where $n = 1$,

$$PV = C \implies P = CV^{-1}$$

$$W = \int_1^2 CV^{-1}\, dV = \boxed{C \ln\left(\frac{V_2}{V_1}\right)}$$

Note that for both cases, the constant C is defined by either state 1 or 2 since by definition of a polytropic process,

$$C = P_1V_1^n = P_2V_2^n$$

so we can rewrite the work of a polytropic process ($n \neq 1$) as

$$W = C\left(\frac{V_2^{-n+1}}{-n+1}\right) - C\left(\frac{V_1^{-n+1}}{-n+1}\right) = P_2V_2^n\left(\frac{V_2^{-n+1}}{-n+1}\right) - P_1V_1^n\left(\frac{V_1^{-n+1}}{-n+1}\right)$$

$$\boxed{W = \frac{P_2\cancel{V}_2 - P_1\cancel{V}_1}{-n+1}}$$

Since the problem doesn't specify that the process is isothermal, we will use the $n \neq 1$ case.

Compression $1 \to 2$ (polytropic)

$$W_{\text{comp}} = \frac{P_2\cancel{V}_2 - P_1\cancel{V}_1}{-n+1}$$

Expansion $2 \to 3$ (ISOBARIC: constant pressure)

$$W_{\text{exp}} = P_2(\cancel{V}_3 - \cancel{V}_2) = P_2(\cancel{V}_1 - \cancel{V}_2)$$

Ratio:

$$\frac{W_{\text{comp}}}{W_{\text{exp}}} = \frac{P_2\cancel{V}_2 - P_1\cancel{V}_1}{(-n+1)(P_2(\cancel{V}_1 - \cancel{V}_2))}$$

Rewriting it using the compression ratio

$$r = \frac{\cancel{V}_1}{\cancel{V}_2} \implies \cancel{V}_1 = r\cancel{V}_2$$

$$P_1\cancel{V}_1^n = P_2\cancel{V}_2^n \quad \text{(polytropic)} \implies P_2 = P_1\left(\frac{\cancel{V}_1}{\cancel{V}_2}\right)^n = P_1 r^n$$

$$\implies \frac{W_{\text{comp}}}{W_{\text{exp}}} = \frac{P_1 r^n \cancel{V}_2 - P_1 r \cancel{V}_2}{(-n+1)(P_1 r^n(r\cancel{V}_2 - \cancel{V}_2))}$$

Cancelling $P_1\cancel{V}_2$ from the numerator and denominator, we get

$$\boxed{\frac{W_{\text{comp}}}{W_{\text{exp}}} = \frac{r^n - r}{(-n+1)r^n(r-1)}}$$

c) Plugging in $n = 1.4$ and $r = 6$:

$$\frac{W_{\text{comp}}}{W_{\text{exp}}} = \frac{6^{1.4} - 6}{(-1.4+1)6^{1.4}(6-1)} \approx \boxed{0.256}$$

4-44

4–44 Two tanks (Tank A and Tank B) are separated by a partition. Initially Tank A contains 2 kg of steam at 1 MPa and 300°C while Tank B contains 3 kg of saturated liquid–vapor mixture at 150°C with a vapor mass fraction of 50 percent. The partition is now removed and the two sides are allowed to mix until mechanical and thermal equilibrium are established. If the pressure at the final state is 300 kPa, determine (*a*) the temperature and quality of the steam (if mixture) at the final state and (*b*) the amount of heat lost from the tanks.

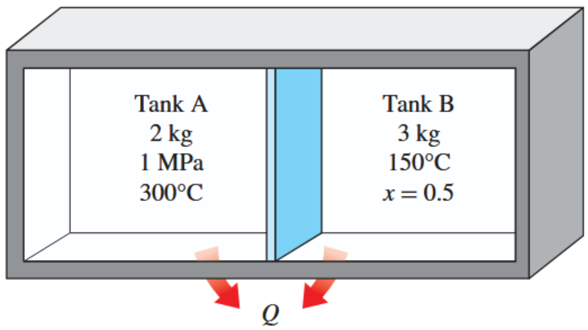


FIGURE P4–44

Given:

- State 1
  - Tank A (STEAM): $m_{A1} = 2$ kg, $P_{A1} = 1$ MPa, $T_{A1} = 300$°C
  - From the superheated water table A-6: $v_{A1} = 0.25799$ m^3/kg, $u_{A1} = 2793.7$ kJ/kg
  - Tank B (SATURATED MIXTURE): $m_{B1} = 3$ kg, $T_{B1} = 150$°C, $x_{B1} = 0.5$
- State 2
  - At mechanical and thermal equilibrium
  - $P_2 = 300$ kPa
- The box is rigid and the two sides mix after being initially separated

Find:

- The temperature and quality of the steam if it is a mixture at state 2
- Amount of heat Q lost during the process

Using the quality of B1, we can find the specific volume and internal energy of B1 at $T_{B1} = 150$°C:

$$v_{B1} = v_f + x_{B1} v_{fg} = [0.001091 + 0.5(0.39248 - 0.001091)] \text{ m}^3/\text{kg} = 0.19696 \text{ m}^3/\text{kg}$$

$$u_{B1} = u_f + x_{B1}u_{fg} = [631.66 + 0.5(1927.4)] \text{ kJ/kg} = 1595.36 \text{ kJ/kg}$$

Now state 1 is fully defined.

a) Since the total volume and mass is conserved, we can find the specific volume at state 2:

$$\cancel{v}_2 = \frac{m_{A1}\cancel{v}_{A1} + m_{B1}\cancel{v}_{B1}}{m_{A1} + m_{B1}} = \frac{[2\,\text{kg}][0.25799\,\text{m}^3/\text{kg}] + [3\,\text{kg}][0.19696\,\text{m}^3/\text{kg}]}{2\,\text{kg} + 3\,\text{kg}} = 0.22137 \text{ m}^3/\text{kg}$$

Using this specific volume and the pressure at state 2, we can determine the phase of state 2: SATURATED MIXTURE since $\cancel{v}_f < \cancel{v}_2 < \cancel{v}_g$ at $P_2 = 300$ kPa.

Now we can find the quality at state 2:

$$x_2 = \frac{\cancel{v}_2 - \cancel{v}_f}{\cancel{v}_g - \cancel{v}_f} = \frac{0.22137 - 0.001073}{0.60582 - 0.001073} = \boxed{0.3642}$$

The temperature at state 2 is the saturation temperature at $P_2 = 300$ kPa, which is $\boxed{133.52°\text{C}}$.

b) Apply the first law for a closed system, keeping in mind that since $\Delta V = 0$ (rigid box), $W = 0$:

$$\Delta U = Q - W$$

$$Q = \Delta U = U_2 - U_1 = m_2 u_2 - (m_{A1}u_{A1} + m_{B1}u_{B1})$$

We can use the quality at state 2 to find the internal energy at state 2:

$$u_2 = u_f + x_2 u_{fg} = 561.11 + 0.3642(1982.1) = 1282.9908 \text{ kJ/kg}$$

Thus,

$$\begin{aligned} Q &= (m_{A1} + m_{B1})u_2 - (m_{A1}u_{A1} + m_{B1}u_{B1}) \\ &= [2 + 3\,\text{kg}][1282.9908\,\text{kJ/kg}] - [2\,\text{kg}][2793.7\,\text{kJ/kg}] - [3\,\text{kg}][1595.36\,\text{kJ/kg}] \\ &= \boxed{-3958.526\,\text{kJ}} \end{aligned}$$

Note that the negative sign indicates that the system lost heat during the process.

4-54

4–54 A mass of 10 g of nitrogen is contained in the spring-loaded piston–cylinder device shown in Fig. P4–54. The spring constant is 1 kN/m, and the piston diameter is 10 cm. When the spring exerts no force against the piston, the nitrogen is at 120 kPa and 27°C. The device is now heated until its volume is 10 percent greater than the original volume. Determine the change in the specific internal energy and enthalpy of the nitrogen. *Answers:* 46.8 kJ/kg, 65.5 kJ/kg

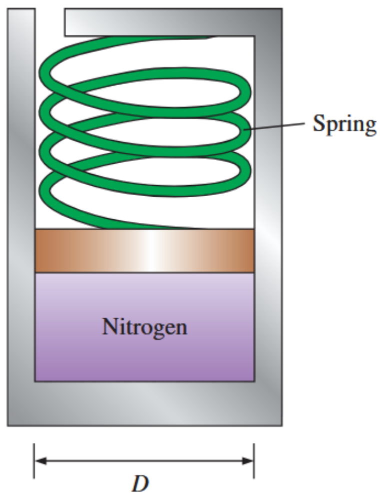


FIGURE P4–54

Given:

- Mass of nitrogen gas $m = 10g$
- $k = 1$ kN/m
- $D = 10$ cm
- State 1: Spring is uncompressed. $P_1 = 120$ kPa, $T_1 = 27°$C, $\cancel{V}_1$
- State 2: $\cancel{V}_2 = 1.1\cancel{V}_1$, P_2, T_2

Find:

- Change in specific internal energy Δu and specific enthalpy Δh from state 1 to state 2

Since the nitrogen is a gas, we can use the ideal gas law:

$$P_1\cancel{V}_1 = mRT_1$$

$$\cancel{V}_1 = \frac{mRT_1}{P_1} = \frac{[0.01\,\mathrm{kg}]\left[0.2968\,\frac{\mathrm{kPa\cdot m^3}}{\mathrm{kg\cdot K}}\right][27 + 273.15\,\mathrm{K}]}{120\,\mathrm{kPa}} = 0.007423\,\mathrm{m}^3$$

$$\cancel{V}_2 = 1.1\cancel{V}_1 = 0.008166\,\mathrm{m}^3$$

Now we can find the displacement of the spring at state 2:

$$\Delta\cancel{V} = \frac{\pi}{4}D^2(x_2 - x_1)$$

$$\implies x_2 = \frac{4\Delta\cancel{V}}{\pi D^2} + x_1 = \frac{4[0.008166 - 0.007423\,\mathrm{m}^3]}{\pi(0.1\,\mathrm{m})^2} + 0 = 0.094601\,\mathrm{m}$$

We can find the force exerted by the spring at state 2 and thus the pressure at state 2:

$$P_2 = P_1 + P_{\mathrm{spring}} = P_1 + \frac{kx_2}{\left(\frac{\pi}{4}D^2\right)}$$

$$P_2 = [120\,\mathrm{kPa}] + \frac{[1000\,\mathrm{N/m}][0.094601\,\mathrm{m}]}{\left(\frac{\pi}{4}(0.1\,\mathrm{m})^2\right)} = [120\,\mathrm{kPa}] + [12044.973\,\mathrm{Pa}]$$

$$P_2 = 132.044\,\text{kPa}$$

Now we can find the temperature at state 2 using the ideal gas law again:

$$T_2 = \frac{P_2 \mathcal{V}_2}{mR} = \frac{[132.044\,\text{kPa}][0.008166\,\text{m}^3]}{[0.01\,\text{kg}]\left[0.2968\,\frac{\text{kPa}\cdot\text{m}^3}{\text{kg}\cdot\text{K}}\right]} = 363.298\,\text{K}$$

Finally, we can find the change in specific internal energy and specific enthalpy using the specific heat constants for nitrogen gas:

$$\Delta u = c_v(T_2 - T_1) = [0.743\,\text{kJ/kg}\cdot\text{K}][363.298 - 300.15\,\text{K}] = \boxed{46.91\,\text{kJ/kg}}$$

$$\Delta h = c_p(T_2 - T_1) = [1.039\,\text{kJ/kg}\cdot\text{K}][363.298 - 300.15\,\text{K}] = \boxed{65.61\,\text{kJ/kg}}$$

4-73E

4–73E A 3-ft^3 adiabatic rigid container is divided into two equal volumes by a thin membrane, as shown in Fig. P4–73E. Initially, one of these chambers is filled with air at 100 psia and 100°F while the other chamber is evacuated. Determine the internal energy change of the air when the membrane is ruptured. Also determine the final air pressure in the container.

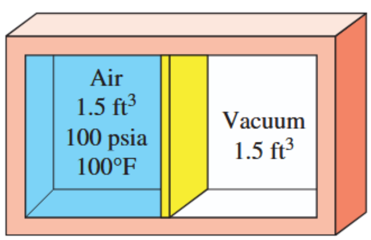


FIGURE P4–73E

Given:

- Total volume $\mathcal{V}_t = 3$ ft^3
- Adiabatic rigid container ($Q = 0$, $W = 0$)
- State 1: separated evenly into two parts by a partition
  - Air: $\mathcal{V}_{A1} = 1.5$ ft^3, $P_{A1} = 100$ psia, $T_{A1} = 100$°F
  - Vacuum: $\mathcal{V}_{V1} = 1.5$ ft^3, $P_{V1} = 0$ psia
- State 2: partition is removed

Find:

- Change in internal energy of the air and the final pressure in state 2.

Using the first law for a closed system, we can find the change in internal energy of the air:

$$\boxed{\Delta U = Q - W = 0}$$

Since the container is rigid and adiabatic, there is no work done and no heat transfer. Therefore, the internal energy of the air remains constant.

To find the pressure at state 2, we can use the ideal gas law. But first we need to find the temperature of air at state 2.

$$u_2 - u_1 = c_v(T_2 - T_1) \implies T_2 = T_1 \quad (\text{since } \Delta U = 0)$$

Thus,

$$T_1 = \frac{P_1 \cancel{V}_1}{mR}, \quad T_2 = \frac{P_2 \cancel{V}_2}{mR}$$

$$T_1 = T_2 \implies P_1 \cancel{V}_1 = P_2 \cancel{V}_2$$

$$P_2 = P_1 \frac{\cancel{V}_1}{\cancel{V}_2} = [100\,\text{psia}] \frac{1.5\,\text{ft}^3}{3\,\text{ft}^3} = \boxed{50\,\text{psia}}$$

4-138

4–138 Two rigid tanks are connected by a valve. Tank A contains 0.2 m³ of water at 400 kPa and 80 percent quality. Tank B contains 0.5 m³ of water at 200 kPa and 250°C. The valve is now opened, and the two tanks eventually come to the same state. Determine the pressure and the amount of heat transfer when the system reaches thermal equilibrium with the surroundings at 25°C. *Answers:* 3.17 kPa, 2170 kJ

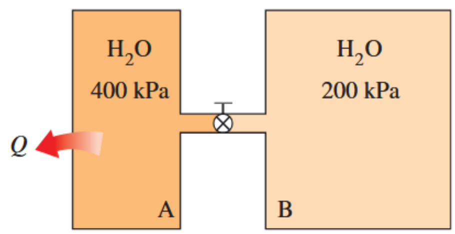


FIGURE P4–138

Given:

- Rigid tank ($\Delta \cancel{V} = 0 \implies W = 0$)
- State 1:
  - A (Water): $\cancel{V}_{A1} = 0.2$ m^3, $P_{A1} = 400$ kPa, $T_{A1} =$?°C, $x_{A1} = 0.8$
  - B (Water): $\cancel{V}_{B1} = 0.5$ m^3, $P_{B1} = 200$ kPa, $T_{B1} =$ 250°C
- State 2: At mechanical and thermal equilibrium. $\cancel{V}_2$, P_2, $T_2 = 25$°C

Find:

- Pressure at state 2 and amount of heat transferred during the process

Start by defining state A1. It is a saturated mixture since $x_{A1} = 0.8$. At 400 kPa, the saturation temperature is $\underline{T_{A1} = 143.61°\text{C}}$.

Using the quality, we can also solve for the specific volume and internal energy at state A1:

$$\cancel{v}_{A1} = \cancel{v}_f + x_{A1} \cancel{v}_{fg} = [0.001084 + 0.8(0.46242 - 0.001084)]\ \text{m}^3/\text{kg} = \underline{0.37015\ \text{m}^3/\text{kg}}$$

$$u_{A1} = u_f + x_{A1} u_{fg} = [604.22 + 0.8(1948.9)]\ \text{kJ/kg} = \underline{2163.34\ \text{kJ/kg}}$$

Next we can define the state B1. At P=200 kPa, $T_{B1} > T_{\text{sat}}$, so it is a superheated vapor. Looking at the superheated vapor tables we find:

$$\underline{\cancel{v}_{B1} = 1.19890\ \text{m}^3/\text{kg}, \quad u_{B1} = 2731.4\ \text{kJ/kg}}$$

With both the total volumes and specific volumes, we can find the masses in state 1:

$$m_{A1} = \frac{\cancel{V}_{A1}}{\cancel{v}_{A1}} = \frac{0.2}{0.37015} = 0.5403 \text{ kg}$$

$$m_{B1} = \frac{\cancel{V}_{B1}}{\cancel{v}_{B1}} = \frac{0.5}{1.19890} = 0.4170 \text{ kg}$$

Now lets define state 2:

$$\cancel{V}_2 = \cancel{V}_{A1} + \cancel{V}_{B1} = 0.2 + 0.5 = 0.7 \text{ m}^3$$

$$\implies \cancel{v}_2 = \frac{\cancel{V}_2}{m_{A1} + m_{B1}} = \frac{0.7}{0.5403 + 0.4170} = 0.7312 \text{ m}^3/\text{kg}$$

At $T_2 = 25°\text{C}$, $\cancel{v}_2$ is between $\cancel{v}_g$ and $\cancel{v}_f$, so state 2 is a saturated mixture. Thus, the pressure at state 2 is the saturation pressure at $T_2 = 25°\text{C}$, which is $\boxed{3.1698 \text{ kPa}}$.

Using the quality at state 2, we can find the internal energy at state 2:

$$x_2 = \frac{\cancel{v}_2 - \cancel{v}_f}{\cancel{v}_g - \cancel{v}_f} = \frac{0.7312 - 0.001003}{43.340 - 0.001003} = 0.01684$$

$$u_2 = u_f + x_2 u_{fg} = 104.83 + 0.01684(2304.3) = 143.653 \text{ kJ/kg}$$

Finally, we can apply the first law to find the amount of heat transferred during the process:

$$\Delta U = Q - W$$

$$Q = U_2 - U_1 = (m_{A1} + m_{B1})u_2 - (m_{A1}u_{A1} + m_{B1}u_{B1})$$

$$Q = (0.5403 + 0.4170)(143.653) - (0.5403)(2163.34) - (0.4170)(2731.4) = \boxed{-2169.910 \text{ kJ}}$$

4-148

4–148 A room contains 75 kg of air at 100 kPa and 15°C. The room has a 250-W refrigerator (the refrigerator consumes 250 W of electricity when running), a 120-W TV, a 1.8-kW electric resistance heater, and a 50-W fan. During a cold winter day, it is observed that the refrigerator, the TV, the fan, and the electric resistance heater are running continuously but the air temperature in the room remains constant. The rate of heat loss from the room that day is
(*a*) 5832 kJ/h (*b*) 6192 kJ/h (*c*) 7560 kJ/h
(*d*) 7632 kJ/h (*e*) 7992 kJ/h

Given:

- $\dot{W}_{\text{refrigerator}} = 250$ W
- $\dot{W}_{\text{TV}} = 120$ W
- $\dot{W}_{\text{heater}} = 1.8$ W
- $\dot{W}_{\text{fan}} = 50$ W
- Refrigerator, TV, fan, and heater are running

Find:

- Rate of heat loss in kJ/h if the temp stays constant

Since temp is constant, the rate of energy entering must equal the rate of energy leaving.

Energy in:

$$\dot{W}_{\text{refrigerator}} + \dot{W}_{\text{TV}} + \dot{W}_{\text{heater}} + \dot{W}_{\text{fan}} = 2220\text{W}$$
$$\implies \dot{W}_{\text{loss}} = 2220\text{W}$$

Convert to kJ/h:

$$\left[2220\,\frac{\text{J}}{\text{s}}\right]\left[\frac{1\text{kJ}}{10^3\text{J}}\right]\left[\frac{3600\text{s}}{1\text{h}}\right] = \boxed{7992\text{ kJ/h (e)}}$$